📡 Network Fundamentals Overview

This combined note brings together key concepts on **VLANs**, **network-management security**, **application-layer protocols**, and **IPv4 NAT**, offering a concise reference for Cisco networking fundamentals.

📚 Content Sources

- Section 1: Explains **VLANs** and related switching concepts from a 53-page PDF document. (source)
- Section 2: Covers **AAA** and **NTP** network-management topics from a 30-page PDF document. (source)
- Section 3: Introduces **Application-layer protocols** and OSI upper layers from a 32-page PDF document. (source)
- Section 4: Details **Network Address Translation (NAT)** for IPv4 from a 43-page PDF document. (source)

Section 1: VLANs in Switched Networks - Switching, Routing, and Wireless Essentials

This section covers VLAN fundamentals, configuration, and trunking from a 53-page PDF document (source)

Module Overview and Objectives

Module Features:

- **Animations** for visual concept demonstrations
- **Videos** introducing skills and ideas
- **Check-Your-Understanding (CYU)** online quizzes
- **Interactive Activities** including simulations
- **Syntax Checker** for Cisco CLI practice
- **PT Activity** Packet-Tracer labs
- **Hands-On Labs** with physical equipment
- **Class Activities** for instructor-led discussions

- **Module Quizzes** for self-assessment

Module Objectives:

- Explain **why VLANs** are used in switched networks
- Describe **frame-forwarding** with multiple switches
- Configure **switch ports** for specific VLANs
- Configure **trunk ports** carrying multiple VLANs
- Configure and verify **Dynamic Trunking Protocol (DTP)**

VLAN Fundamentals

> A **VLAN (Virtual LAN)** is a logical grouping of devices that act as if on the same physical LAN

Key Characteristics:

- Segments devices into **broadcast domains**
- Isolates **broadcasts, multicasts, and unicasts** per VLAN
- Each VLAN uses its own **IP-addressing scheme**
- Reduces **broadcast traffic** → smaller broadcast domains

Benefits of VLAN Design:

Benefit	Description
Smaller Broadcast Domains	Less unnecessary traffic, faster convergence
Improved Security	Only same-VLAN devices communicate directly
Improved IT Efficiency	Group by function (faculty vs. students)
Reduced Cost	One switch hosts multiple logical networks
Better Performance	Less broadcast congestion
Simpler Management	Uniform policies for similar devices

Types of VLANs:

VLAN Type	Primary Role	Notes

Default VLAN (VLAN 1)	Default data, native, management	Cannot be deleted/renamed
Data VLAN	User traffic (email, web)	VLAN 1 is default
Native VLAN	Untagged VLAN on 802.1Q trunk	Usually VLAN 1
Management VLAN	Switch management traffic	SVI on Layer-2 switch
Voice VLAN	VoIP traffic	Requires QoS, <150ms latency

VLANs in Multi-Switched Environment

Trunk Definition:

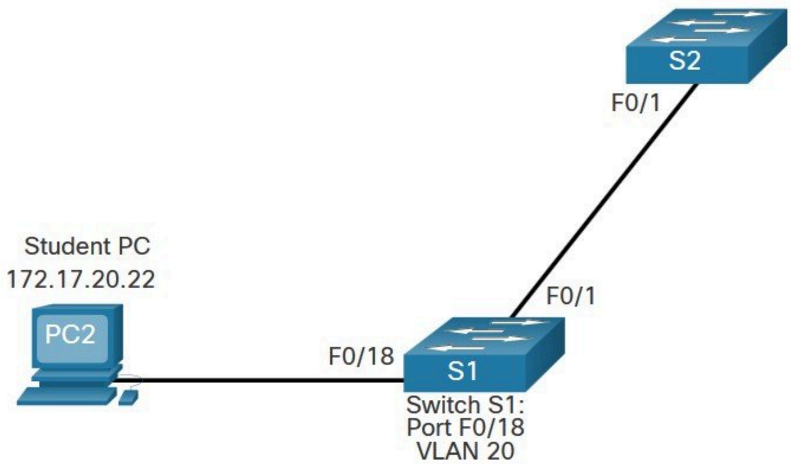
- Point-to-point link carrying **multiple VLAN traffic**
- Uses **802.1Q tagging** by default
- Allows **all VLANs** by default

802.1Q Tag Structure:

Field	Size	Description
TPID	2 bytes	Fixed value 0x8100
User Priority	3 bits	Class of Service (CoS)
CFI	1 bit	Token-ring compatibility
VLAN ID	12 bits	Up to 4096 VLANs (0,4095 reserved)

Native VLAN Rules:

- Only VLAN that is **untagged** on trunk
- Default = **VLAN 1**
- Both ends must agree or traffic may drop



VLAN Configuration

VLAN Ranges on Catalyst Switches:

Range	VLAN IDs	Use
Normal	1-1005	Small-medium LANs
Extended	1006-4095	Larger deployments
Reserved	1002-1005	Legacy VLANs
Default	1	Auto-created

Creating a VLAN:

```
Switch# configure terminal
Switch(config)# vlan 20
Switch(config-vlan)# name student
Switch(config-vlan)# end
```

Assigning Port to VLAN:

```
Switch(config)# interface fa0/18
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
```

Data + Voice VLAN on Single Port:

- Access port carries **one data VLAN** and **one voice VLAN**
- Voice VLAN: switchport voice vlan 150
- QoS trust: mls qos trust cos

```
S1(config)# interface fa0/18
S1(config-if)# no switchport access vlan
S1(config-if)# end
S1#
S1# show vlan brief
VLAN Name                  Status    Ports
---- ------------------ --------- ------------------------------
1    default            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                  Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                  Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                  Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                  Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                  Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                  Gi0/1, Gi0/2
20   student            active
1002 fddi-default       act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default    act/unsup
1005 trnet-default      act/unsup
```

Verification Commands:

- show vlan brief
- show vlan id 20
- show interface fa0/18 switchport

VLAN Trunks

Trunk Configuration:

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk native vlan 99
Switch(config-if)# switchport trunk allowed vlan 10,20,30,99
```

Verification:

- show interface fa0/1 switchport
- Displays mode, native VLAN, allowed VLANs

Dynamic Trunking Protocol (DTP)

DTP Modes:

Mode	Behavior

access	Forced access mode
dynamic auto	Passive - becomes trunk if requested
dynamic desirable	Active - attempts trunk formation
trunk	Permanent trunk

Best Practice: Disable DTP with switchport nonegotiate

Section 2: Network Management - AAA and NTP Configuration

This section covers network management focusing on AAA and NTP from a 30-page PDF document (source)

AAA Core Concepts

AAA Components:

Component	Definition	Sequence
Authentication	Identifies user before access	First step
Authorization	Determines user permissions	After authentication
Accounting	Records user activity	After authentication

AAA Benefits:

- **Flexibility & Control** - fine-grained authorization
- **Scalability** - central management
- **Standardized Methods** - RADIUS protocol
- **Redundancy** - multiple server groups

AAA Protocols: RADIUS vs TACACS+

Feature	RADIUS	TACACS+
Transport	UDP (1812/1813)	TCP (49)

Authentication	Single-direction	Two-way
Vendor	Open standard	Cisco-proprietary
Use Cases	User-to-network access	Command-level authorization

Configuring AAA on Cisco Switches

Enable AAA:

```
Switch(config)# aaa new-model
```

RADIUS Configuration:

```
Switch(config)# radius server
Switch(config-radius-server)# address ipv4
Switch(config-radius-server)# key
```

TACACS+ Configuration:

```
Switch(config)# tacacs server
Switch(config-server-tacacs)# address ipv4
Switch(config-server-tacacs)# key
```

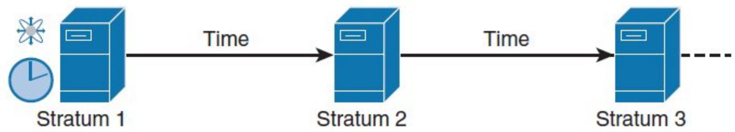
Network Time Protocol (NTP)

Why Accurate Time Matters:

- **Log correlation**
- **Security timestamps**
- **Script scheduling**
- **Troubleshooting**

NTP Stratum Concept:

- **Stratum 0**: Reference clock (atomic)
- **Stratum 1**: Directly attached to Stratum 0
- **Stratum 2+**: Additional hops away



NTP Operational Modes:

- **Server** - supplies time
- **Client** - pulls time
- **Peers** - bidirectional exchange
- **Broadcast/Multicast** - push mode

Verification Commands:

- show ntp status
- show ntp associations

```
R1# show ntp status
Clock is synchronized, stratum 2, reference is 209.165.200.187
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**10
ntp uptime is 1500 (1/100 of seconds), resolution is 4000
reference time is D67E670B.0B020C68 (05:22:19.043 PST Mon Jan 13 2014)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 630.22 msec, peer dispersion is 189.47 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is 0.000000000 s/s
system poll interval is 64, last update was 5 sec ago.
```

Section 3: Application Layer Protocols - Network Fundamentals

This section explores application layer protocols including HTTP, DNS, DHCP, and file sharing from a 32-page PDF document (source)

Application Layer Overview

TCP/IP Application Layer Protocols:

Service	Protocol	Transport	Port(s)	Function
Name System	DNS	TCP/UDP	53	Domain to IP translation
Host Config	DHCP	UDP	67/68	Dynamic IP assignment

Web	HTTP	TCP	80/8080	Web content exchange
Email	SMTP/POP/IMAP	TCP	25/110/143	Mail services
File Transfer	FTP	TCP	20/21	File transfer
File Sharing	SMB	TCP	445	Network file sharing

Peer-to-Peer Networks

P2P Characteristics:

- Any device can be **client and server** simultaneously
- Roles assigned **per request**
- Common applications: **BitTorrent, eDonkey, Freenet**

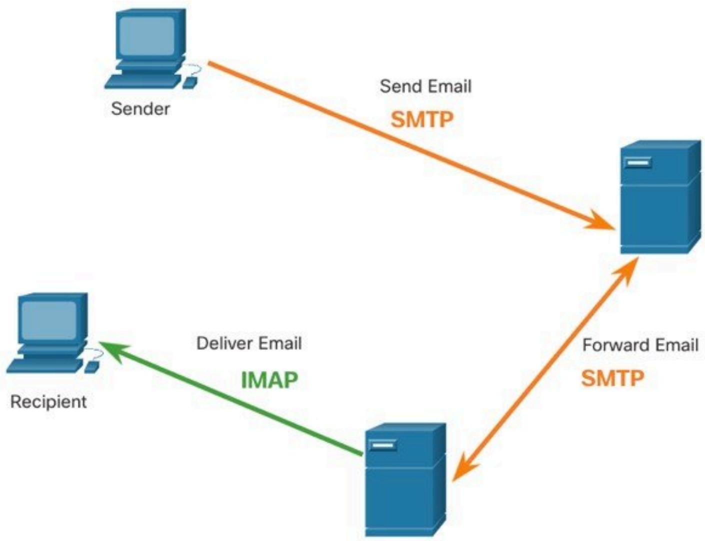
Web and Email Protocols

HTTP Workflow:

1. **URL Parsing** - scheme, host, path
2. **DNS Resolution** - name to IP
3. **GET Request** - request page
4. **Response** - server returns content

Email Protocols:

- **SMTP** (TCP 25) - **sending** mail
- **POP** (TCP 110) - **retrieving** mail (download-delete)
- **IMAP** (TCP 143) - **retrieving** mail (keep on server)



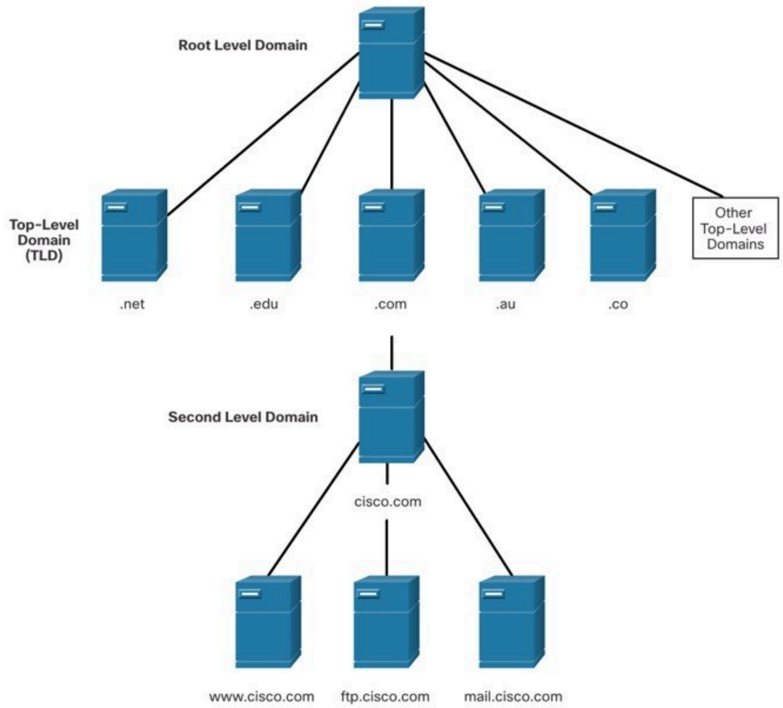
DNS and DHCP Services

DNS Hierarchy:

- **Root Level** - "."
- **TLDs** - .com, .org
- **Second-Level** - cisco.com
- **Subdomains** - www.cisco.com

DNS Record Types:

- **A** - IPv4 address
- **AAAA** - IPv6 address
- **NS** - Name server
- **MX** - Mail exchange



DHCP Process:

1. **DHCPDISCOVER** - client finds server
2. **DHCPOFFER** - server offers IP
3. **DHCPREQUEST** - client requests lease
4. **DHCPACK** - server acknowledges

File Sharing Services

FTP Mechanics:

- **Control Connection** - TCP 21 for commands
- **Data Connection** - TCP 20 for transfers
- Supports **download and upload**

SMB Protocol:

- Client-server request-response
- Functions: **Session management, Resource control, Application messaging**

Section 4: Network Address Translation for IPv4

This section details NAT types, configuration, and troubleshooting from a 43-page PDF document ([source]())

NAT Overview

> **NAT (Network Address Translation)** translates private IPv4 addresses to public addresses, conserving scarce public address space

NAT Terminology:

Term	Description
Inside local	Private IP before translation
Inside global	Public IP after translation
Outside local	External private IP (rare)
Outside global	External public IP

Types of NAT

1. Static NAT:

- **One-to-one** mapping
- **Manually configured**
- Use case: **inbound server access**

2. Dynamic NAT:

- **Pool of public addresses**
- **First-come, first-served**
- Requires **enough public IPs**

Dynamic NAT Configuration Steps	
Step 1	Define a pool of global addresses to be used for translation. **ip nat pool** *name start-ip end-ip* {**netmask** *netmask* | **prefix-length** *prefix-length*}
Step 2	Configure a standard access list permitting the addresses that should be translated. **access-list** *access-list-number* **permit** *source* [*source-wildcard*]
Step 3	Establish dynamic source translation, specifying the access list and pool defined in prior steps. **ip nat inside source list** *access-list-number* **pool** *name*
Step 4	Identify the inside interface. **interface** *type number* **ip nat inside**
Step 5	Identify the outside interface. **interface** *type number* **ip nat outside**

Dynamic NAT Configuration:

```
ip nat pool MYPOOL 203.0.113.10 203.0.113.20 netmask 255.255.255.0
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 pool MYPOOL
interface GigabitEthernet0/0
 ip nat inside
interface GigabitEthernet0/1
 ip nat outside
```

3. PAT (Port Address Translation):

- **Many-to-one** mapping
- Uses **source ports** for differentiation
- Most **address-efficient** method

NAT Benefits and Drawbacks

Benefits:

- Conserves **public address space**
- Adds **flexibility** for internal networks
- Simplifies **internal address planning**
- Improves **security** (hides internal topology)

Drawbacks:

- **Performance impact** - translation overhead
- **Reduced end-to-end functionality**
- **Loss of IP traceability**
- **Complicated tunneling** for VPNs

NAT Configuration Examples

Static NAT:

```
ip nat inside source static 192.168.1.10 203.0.113.10
```

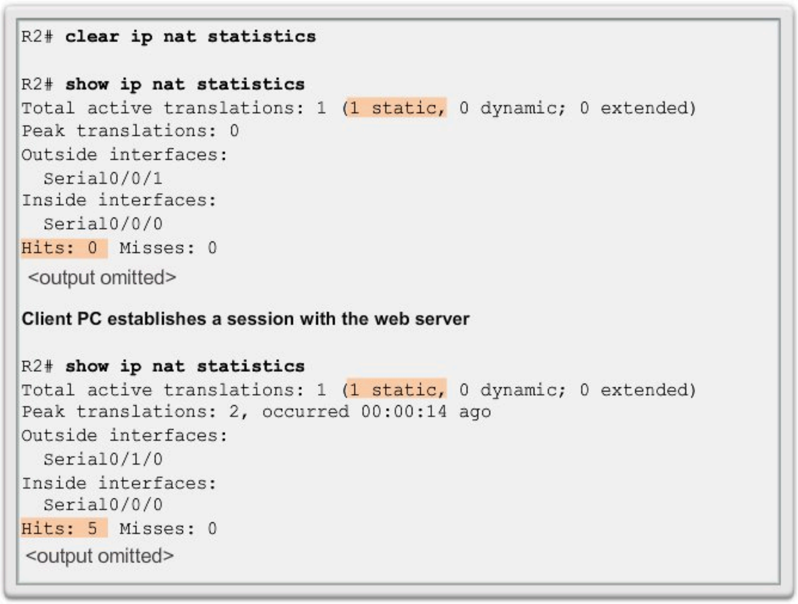
PAT Overload:

```
ip nat inside source list 1 interface GigabitEthernet0/1 overload
```

NAT Verification and Troubleshooting

Verification Commands:

- show ip nat translations - view mappings
- show ip nat statistics - view counters
- debug ip nat - real-time debugging



```
R2# clear ip nat statistics

R2# show ip nat statistics
Total active translations: 1 (1 static, 0 dynamic; 0 extended)
Peak translations: 0
Outside interfaces:
  Serial0/0/1
Inside interfaces:
  Serial0/0/0
Hits: 0  Misses: 0
 <output omitted>

Client PC establishes a session with the web server

R2# show ip nat statistics
Total active translations: 1 (1 static, 0 dynamic; 0 extended)
Peak translations: 2, occurred 00:00:14 ago
Outside interfaces:
  Serial0/1/0
Inside interfaces:
  Serial0/0/0
Hits: 5  Misses: 0
 <output omitted>
```

Port Forwarding:

```
ip nat inside source static tcp 192.168.1.100 80 203.0.113.10 80
```

🎯 Putting it All Together

🌐 Integrated Network Blueprint

- **Segment** devices with **VLANs** to create isolated broadcast domains, then attach those VLANs to **router-on-a-stick** or Layer-3 switches for inter-VLAN routing.
- Secure every access point (console, VTY, Ethernet) using **AAA**:
    1. Enable the AAA framework.

   2. Define **RADIUS** (or **TACACS+**) server groups.
   3. Apply **authentication**, **authorization**, and **accounting** method lists to the appropriate lines and interfaces.
- Keep **time consistent** across all devices with **NTP**; configure each switch/router as an NTP client of a reliable stratum-1 or stratum-2 source, then verify with show ntp status and show ntp associations.

> *VLAN – a logical grouping of ports that behave as if they share a single LAN, reducing broadcast traffic and enhancing security.*

> *AAA – the three-step security model (Authentication → Authorization → Accounting) that centralizes user control and audit trails.*

> *NTP – a protocol that synchronizes clocks across the network, essential for accurate logging and time-based security policies.*

📡 Application-Layer Services on the Segmented Fabric

Service	Typical Protocol	Transport	Port(s)	Placement in the Design
Web browsing	HTTP / HTTPS	TCP	80 / 443	Served from a **DMZ VLAN**; ACLs restrict access to the internal VLANs.
Email	SMTP, POP, IMAP	TCP	25 / 110 / 143	Mail servers reside in a **dedicated VLAN** with **AAA-controlled** admin access.
Name resolution	DNS	UDP/TCP	53	DNS forwarders placed in a **service VLAN**; NTP-synchronize for accurate TTL handling.

File sharing	FTP, SMB	TCP	21 / 20, 445	Hosted in a **data VLAN**; access governed by **authorization** profiles.
VoIP	SIP, RTP	UDP/TCP	5060, dynamic	Placed in a **voice VLAN** with QoS trusted on the access ports.

> *Application layer – the OSI layer that enables end-user programs to exchange data, often relying on underlying transport-layer protocols for reliability.*

🔀 NAT & PAT for Edge Connectivity

- **Static NAT** exposes services (e.g., a web server in the DMZ) by mapping a fixed **inside-local** address to an **inside-global** address.
- **Dynamic NAT** draws from a public pool when internal hosts need outbound Internet access, ensuring each session gets a unique public address.
- **PAT (NAT overload)** lets the entire inside network share a single public IP, differentiating flows by source ports; this is the default for most campus edge routers.

NAT Type	Mapping	Public-IP consumption	Typical use
Static	1:1	One per service	Inbound access to servers
Dynamic	1:1 (temporary)	One per simultaneous session	Bulk outbound traffic when enough public IPs are available
PAT	Many-to-1 (ports)	One (or few)	General Internet access for all users

- **Port forwarding** extends static NAT with a specific port number, enabling external clients to reach an internal service (e.g., ip nat inside source static tcp 192.168.10.10 80 203.0.113.5 80).

🛠️ Practical Workflow (Step-by-Step)

1. **Design VLAN topology** → assign VLAN IDs, create trunks, set native VLANs.
2. **Apply AAA** → configure server groups, method lists, and bind them to VTY/console lines.
3. **Synchronize clocks** → set NTP source, verify stratum and offset.
4. **Deploy services** → place servers in appropriate VLANs, configure ACLs, and enable required QoS.
5. **Configure NAT/PAT** → choose static, dynamic, or overload based on traffic patterns; add any necessary port-forwarding entries.
6. **Validate** → use show vlan brief, show ip nat translations, show ntp status, and show running-config | include aaa to confirm each layer functions as intended.

📊 Cross-Layer Benefits at a Glance

Layer	Goal	How the combined features help
Layer 2 (VLAN)	Isolation & broadcast control	Limits unnecessary traffic, reduces attack surface.
Layer 3 (Routing/NAT)	Connectivity to external networks	NAT conserves public addresses; routing enables inter-VLAN communication.
Layer 4 (Transport)	Reliable session handling	PAT tracks ports, ensuring return traffic finds the right host.
Layer 7 (Application)	Service delivery	AAA restricts who can use services; NTP guarantees consistent timestamps for logs and security events.

> *A cohesive network stitches together logical segmentation, robust security, precise timekeeping, and efficient address translation—delivering reliable, secure, and manageable services across the campus.*